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Assignment 202

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IARCO RESEARCH PROPOSAL

AI-Driven Near Real-Time Monitoring of Unsanctioned Sand Mining and Riverbank Erosion Vulnerability Using Sentinel-1 SAR Imagery and Geospatial Analysis

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Research Problem

Bangladesh's rivers are constantly shifting due to strong hydrological activity and it causes bank erosion. This problem is getting worse by unregulated sand extraction practices, which change river morphology and accelerate erosion, posing serious threats to agriculture, habitation, and local livelihoods. The need for rapid and effective detection of such alterations is critical; optical satellites, such as Landsat, provide valuable data for land and water monitoring, but require careful analysis to yield meaningful land cover insights.

Cloud cover presents a challenge to optical sensors but can be overcome with the use of synthetic aperture radar (SAR). Although SAR has advantages when acquiring images through dense cloud cover, day or night, this data cannot establish illegal sand mining on its own. Changes in water levels, boat movement, and moving sandbars alter SAR image reflections. By using machine learning models to multi-temporal SAR data, we can spot potential illegal sand mining more effectively.

In this study, we focus on a simple and practical way to track sand mining and see how it affects nearby riverbanks. We combine Sentinel-1 SAR imagery with random forest classification and shoreline mapping to spot potential mining areas and erosion-prone zones. This model works as a decision-support tool rather than an official proof of illegal activity.

Existing Literature

Researchers in remote sensing often use optical indices such as NDWI and MNDWI to classify different land cover types [1]. Google Earth Engine (GEE) makes it easy to process these large satellite datasets in the cloud [2]. In regions where the optical sensor's view is frequently prevented by clouds, Sentinel-1 C-band SAR offers a helpful alternative [3].

Random Forest (RF) is a well-known and well-regarded machine learning tool for classifying satellite images [4]. Using SAR backscatter values with spatial texture features helps to separate land cover classes more clearly. To analyze the shifting riverbank morphology across time, geospatial shoreline tools take measurements from a defined set of transect lines [5]. Previous studies have shown that, utilizing Sentinel-1 data works well for analyzing riverbank erosion in Bangladesh [6]. Other studies have also examined the application of remote sensing and geospatial analysis for analyzing shorelines and river morphology across the region in Bangladesh [7], [8].

Still, the combination of time-series Sentinel-1 images, machine learning algorithms, anomaly detection, and geospatial shoreline analysis to detect potential illegal sand mining and erosion-biased zones is a hopeful area of research.

Research Question

How effectively can Sentinel-1 SAR dual-polarization imagery, paired with random forest classification and shoreline mapping, detect unsanctioned sand mining and assess riverbank erosion during Bangladesh's monsoon season?

Objectives

1. Classify major river-surface and bank zones using Sentinel-1 SAR data features.
2. Detect spatio-temporal anomalies that point toward potential unsanctioned sand mining activity
3. Measure riverbank movement over time and build a prototype alert system for high-risk erosion zones

Methodology

To execute this study, we will process Sentinel-1 SAR datasets and perform geospatial mapping using GEE, using Python scripts for data processing.

Data Preparation: We will collect Sentinel-1 GRD IW images in both VV and VH polarizations. We will apply standard speckle filtering and perform the necessary terrain corrections to improve image quality.

Feature Extraction: We will extract the backscatter values of VV and VH, as well as x-polarization ratios and spatial texture features (such as contrast, entropy, correlation, and variance). These help distinguish deep water, sandbars, sediment deposits, and vegetated banks.

Machine learning classification: We will use a random forest model to classify the study area into water, active channels, sandbars, and riverbanks. The training and validation samples will be kept separate.

Probable mining detection: Instead of marking every small radar change, we look at how changes build up over time across multiple images. Places that show continuous shifts in exposed sand or river channel features will be flagged as potential mining spots.

Riverbank analysis: We will extract water-land boundaries from the classified images. Transects will measure changes in position over time. Net movement and shifting rates will show erosion trends and highlight high-risk areas.

Accuracy assessment: To see how well the model works, we will check classification accuracy using a confusion matrix along with standard measures like overall accuracy, precision, recall, F1-score, and Kappa.

Alert prototype: The system creates a map-based monitoring tool that shows the location, observation date, detected change, and confidence level. It works as a decision-support prototype and gives authorities actionable data monitoring.

Ethical considerations: This research will use satellite and geospatial information and will not collect personal information. The detected locations will be reported as probable activities and should be independently verified before legal or law enforcement action.

Research Topic

This framework leverages the power of Sentinel-1 SAR images, machine learning and geospatial mapping to monitor sand extraction and assess erosion risks. The main contribution of this framework is that, it has the ability to provide repeated monitoring where optical satellites fail due to cloud cover.

The system provides valuable data for local authorities, enabling them to localize high-risk zones and conduct field checks while regular satellite monitoring is cost-effective for large river basins.

Limitations: SAR backscatter can be affected by water level, vegetation, and sediment flow. Thus, radar anomalies cannot confirm illegal sand extraction. So we recommend Ground checks and field observations to confirm the conclusions drawn from remote sensing.

Quality of Writing

The methodology directly answers the research question, and we carefully choose our words to separate suspected sand mining from officially confirmed illegal activity. References follow the IEEE style.

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